

SHLOK ANAND

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Smart Resume Data Extractor

You are asked to build a small JavaScript program that processes raw resume text (copied from a plain .txt file). Since resumes are often messy (extra spaces, inconsistent capitalization, multiple lines, etc.), your program must **clean, validate, and extract key information** using **string functions and regular expressions**.

Input (raw text string example):

```
let resume = `
```

```
Name: jOhN doE
```

```
Email: john.doe@Example.com
```

```
Phone: 9876543210
```

```
Skills: HTML, CSS, JavaScript, Python, SQL
```

```
Extra: Contact me at john.altmail@example.org
```

```
#developer #coder
```

```
`;
```

Expected Output (after processing):

```
{
  name: "John Doe", // Trimmed & formatted correctly
  email: "john.doe@example.com", // Validated and lowercased
  phone: "9876543210", // Validated mobile number
  skills: ["HTML", "CSS", "JavaScript", "Python", "SQL"], // Clean split
  hashtags: ["#developer", "#coder"], // Extracted using regex
  emailCount: 2, // Count of email addresses found
  wordCount: 15, // Count words in resume text
  vowelCount: 24 // Count vowels in resume text
}
```

Steps

Students must implement **all the following sub-tasks** in one program:

1. Name Formatting

Extract the name, remove extra spaces (trim() + replace()),

Capitalize the first letter of each word (charAt() + slice()).

2. Email Validation & Extraction

Use regex to find *all* email addresses.

Store the first one as the primary email.

Count how many emails were found.

3. Mobile Number Validation

Validate that the phone number starts with 6,7,8,9 and has exactly 10 digits.

If invalid, output "Invalid phone number".

4. Skills Processing

Extract skills from the line.

Split by comma, trim spaces, and return as an array.

5. Hashtag Extraction

Extract all hashtags (e.g., #developer, #coder).

6. Word Count

Count total number of words in the resume.

7. Vowel Count

Count total vowels (a, e, i, o, u).

The following code contains function names + comments only, so you have to fill in the logic yourself (using what you have learned in Practical 6 Part 1 & 2).

// Final Project: Smart Resume Data Extractor

```
let resume = `
```

```
Name: jOhN doE
```

```
Email: john.doe@Example.com
```

```
Phone: 9876543210
```

```
Skills: HTML, CSS, JavaScript, Python, SQL
```

```
Extra: Contact me at john.altmail@example.org
#developer #coder
`;
// 1. Format Name
function formatName(rawName) {
  // TODO: Trim spaces, split into words, capitalize each word properly
  return formattedName;
}

// 2. Extract and Validate Emails
function extractEmails(text) {
  // TODO: Use regex to find all emails
  // Return array of emails
}

// 3. Validate Mobile Number
function validateMobile(number) {
  // TODO: Check if starts with 6/7/8/9 and has exactly 10 digits
  // Return valid number or "Invalid phone number"
}

// 4. Process Skills
function processSkills(rawSkills) {
  // TODO: Split by comma, trim each skill, return array
}

// 5. Extract Hashtags
function extractHashtags(text) {
  // TODO: Use regex to find all hashtags (#...)
  // Return array of hashtags
}

// 6. Word Count
function countWords(text) {
  // TODO: Split by spaces, filter empty items, return count
}

// 7. Vowel Count
```

```

function countVowels(text) {
  // TODO: Loop through text and count vowels (a, e, i, o, u)
}
// -----
// Main Program
// -----

function processResume(resumeText) {
let output = {};

  // Extract Name
  // (Hint: use regex or string search for "Name:")
  // output.name = formatName(...);

  // Extract Emails
  // let emails = extractEmails(resumeText); //
  output.email = emails[0]; // Primary email //
  output.emailCount = emails.length;

  // Extract Phone
  // (Hint: find line with "Phone:")
  // output.phone = validateMobile(...);

  // Extract Skills
  // (Hint: find line with "Skills:")
  // output.skills = processSkills(...);

  // Extract Hashtags
  // output.hashtags = extractHashtags(resumeText);

  // Count Words
  // output.wordCount = countWords(resumeText);

  // Count Vowels
  // output.vowelCount = countVowels(resumeText);

  return output;
}

```

```
// Run Program
console.log(processResume(resume));
```

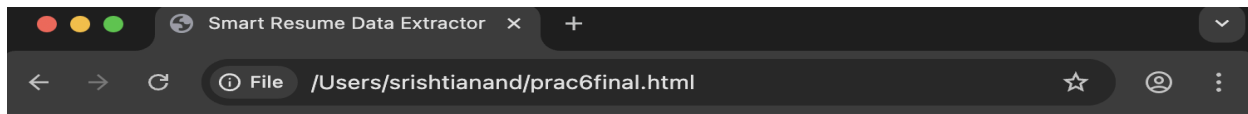
Test Input 1 (Normal, Clean Example)

```
let resume1 = `
Name: jOhN doE
Email: john.doe@Example.com
Phone: 9876543210

Skills: HTML, CSS, JavaScript, Python, SQL

Extra: Contact me at john.altmail@example.org
#developer #coder
`;
```

Expected: Properly formatted name, 2 emails, valid phone, 5 skills, 2 hashtags, correct word & vowel count.



Smart Resume Data Extractor

Paste your raw resume text below:

```
let resume1 = `
Name: j0hN doE
Email: john.doe@Example.com
Phone: 9876543210
Skills: HTML, CSS, JavaScript, Python, SQL
Extra: Contact me at john.altmail@example.org
#developer #coder
`;
```

Process Resume

Extracted Output:

```
{
  "name": "John Doe",
  "email": "john.doe@example.com",
  "emailCount": 2,
  "phone": "9876543210",
  "skills": "HTML, CSS, JavaScript, Python, SQL",
  "hashtags": "#developer #coder",
  "wordCount": 25,
  "vowelCount": 46
}
```

Test Input 2 (Messy Spaces & Invalid Phone)

```
let resume2 = `
```

Name: aLIcE SMiTH

Email: alice.smith@work.com

Phone: 1234567890

Skills: C, C++ ,Java, JavaScript

Extra Info: Reach me also at alice.backup@mail.org

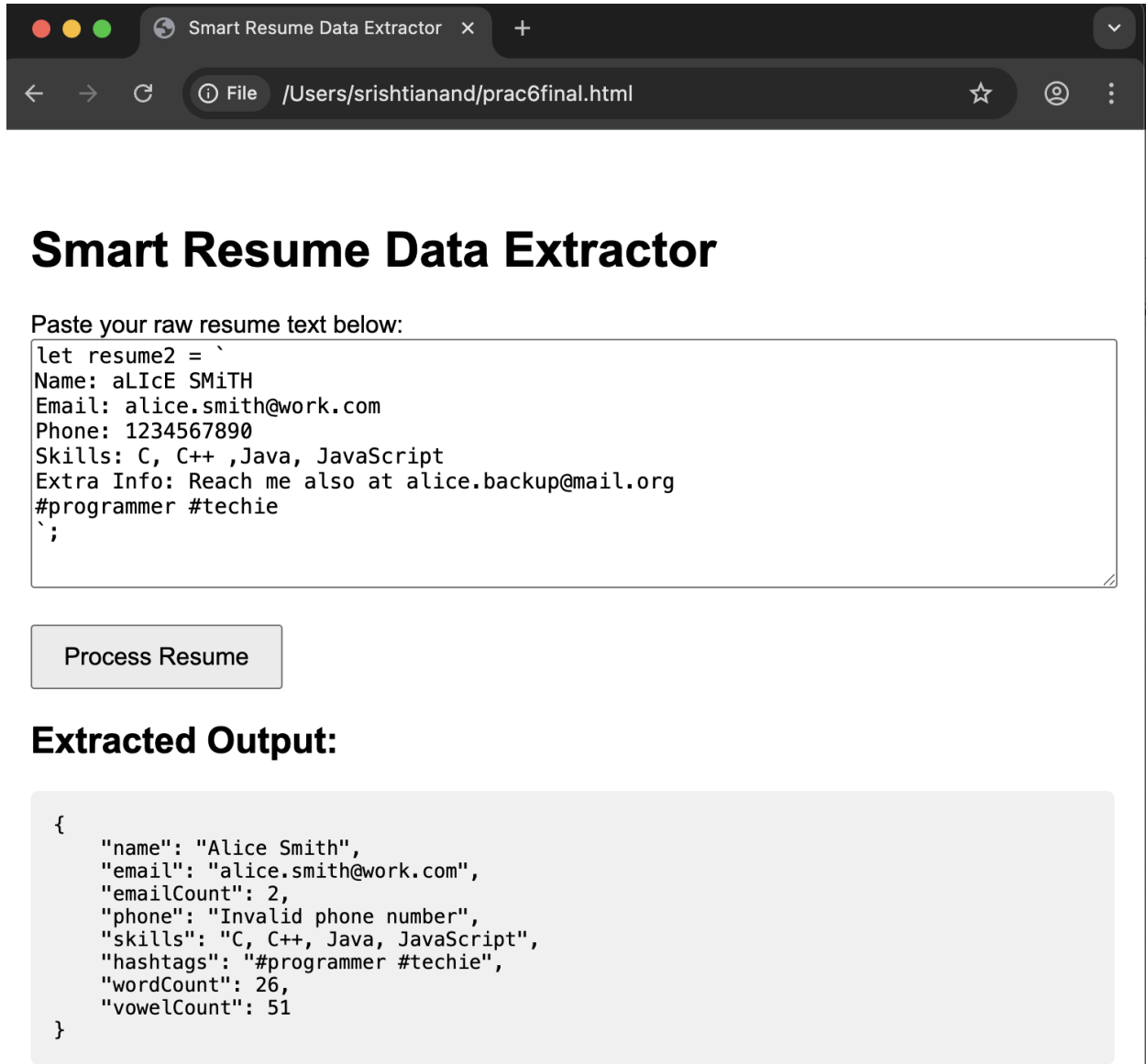
#programmer #techie

```
`;
```

Expected:

- Name → Alice Smith

- 2 emails extracted
- Phone → "Invalid phone number" (since starts with 1)
- Skills → ["C", "C++", "Java", "JavaScript"]
- Hashtags → ["#programmer", "#techie"]



Smart Resume Data Extractor

Paste your raw resume text below:

```
let resume2 = `
Name: aLIcE SMiTH
Email: alice.smith@work.com
Phone: 1234567890
Skills: C, C++ ,Java, JavaScript
Extra Info: Reach me also at alice.backup@mail.org
#programmer #techie
`;
```

Process Resume

Extracted Output:

```
{
  "name": "Alice Smith",
  "email": "alice.smith@work.com",
  "emailCount": 2,
  "phone": "Invalid phone number",
  "skills": "C, C++, Java, JavaScript",
  "hashtags": "#programmer #techie",
  "wordCount": 26,
  "vowelCount": 51
}
```

Test Input 3 (Multiple Emails & Extra Hashtags)

```
let resume3 = `
Name: ROBERT broWn
Email: robert_brown@example.com
```

Phone: 8765432109

Skills: PHP,MySQL, HTML ,CSS, JavaScript

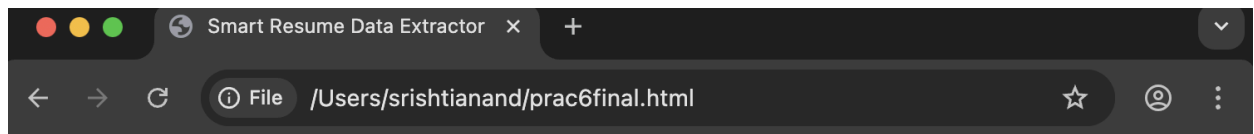
Other: robert.personal@domain.org robert.dev@github.com

#fullstack #developer #javascript #100DaysOfCode

`;

Expected:

- Name → Robert Brown
- 3 emails extracted
- Valid phone
- Skills → ["PHP", "MySQL", "HTML", "CSS", "JavaScript"]
- 4 hashtags extracted



Smart Resume Data Extractor

Paste your raw resume text below:

```
let resume3 = `  
Name: ROBERT brown  
Email: robert_brown@example.com  
Phone: 8765432109  
Skills: PHP,MySQL, HTML ,CSS, JavaScript  
Other: robert.personal@domain.org robert.dev@github.com #fullstack #developer  
#javascript #100DaysOfCode  
`;  
`;
```

Process Resume

Extracted Output:

```
{  
  "name": "Robert Brown",  
  "email": "robert_brown@example.com",  
  "emailCount": 3,  
  "phone": "8765432109",  
  "skills": "PHP, MySQL, HTML, CSS, JavaScript",  
  "hashtags": "#fullstack #developer #javascript #100DaysOfCode",  
  "wordCount": 24,  
  "vowelCount": 55  
}
```


Test Input 4 (Extra Noise, Mixed Case, Extra Spaces)

let resume4 = `

Random Text: Ignore this line

Name: keVin o'CONNOR

Email: kevin.oConnor@Mail.com

Phone: 7894561230

Skills: Python, Machine Learning, Data Science , AI

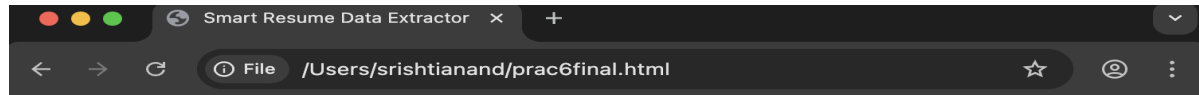
Extra Notes: You can also contact: kevin.backup123@mail.org

#AI #MachineLearning #DataScience #coder

`;

Expected:

- Name → Kevin O'Connor (handle apostrophe properly)
- 2 emails extracted
- Valid phone
- Skills → ["Python", "Machine Learning", "Data Science", "AI"] ●
- Hashtags → ["#AI", "#MachineLearning", "#DataScience", "#coder"]



Smart Resume Data Extractor

Paste your raw resume text below:

```
let resume4 = `
Random Text: Ignore this line
Name: keVin o'CONNOR
Email: kevin.oConnor@Mail.com
Phone: 7894561230
Skills: Python, Machine Learning, Data Science , AI
Extra Notes: You can also contact: kevin.backup123@mail.org #AI #MachineLearning
#DataScience #coder
`;
```

Process Resume

Extracted Output:

```
{
  "name": "Kevin O'Connor",
  "email": "kevin.oconnor@mail.com",
  "emailCount": 2,
  "phone": "7894561230",
  "skills": "Python, Machine Learning, Data Science, AI",
  "hashtags": "#AI #MachineLearning #DataScience #coder",
  "wordCount": 36,
  "vowelCount": 81
}
```